

GD2-CAR–Engineered Microglia Exhibit Antitumor Effects in Organoids and Animal Models of Retinoblastoma

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Received: July 14, 2025

Accepted: October 15, 2025

Published: November 13, 2025

Citation: Xu J, Gao Y, Wang J, et al. GD2-CAR–Engineered microglia exhibit antitumor effects in organoids and animal models of retinoblastoma. *Invest Ophthalmol Vis Sci*. 2025;66(14):26.

<https://doi.org/10.1167/iovs.66.14.26>

PURPOSE. Retinoblastoma (RB) is the most prevalent intraocular malignancy in children, which significantly impacts patients' quality of life. Chimeric antigen receptor (CAR) immune cell therapies opened a door to treating tumors. However, whether microglia, the resident immune cell in the retina, could be engineered to treat retinoblastoma remains unknown. The purpose of this study is to generate human CAR-microglia and to investigate the antitumor effects of CAR-microglia in RB.

METHODS. The presence of CAR in microglia cells was verified using a fluorescence microscope, PCR, and flow cytometry analysis. The CAR-microglia generated from induced pluripotent stem cells were identified by immunofluorescence, flow cytometry analysis, and RNA sequencing (RNA-seq). The antitumor effects of CAR-microglia in retinoblastoma are investigated in vitro and in vivo by live imaging, flow cytometry analysis, RNA-seq, quantitative real-time PCR, cell viability analysis, cell bioluminescence analysis, ELISA, optical coherence tomography, fundus photography, bioluminescence imaging, and hematoxylin and eosin staining.

RESULTS. We developed human GD2 CAR-microglia and found that they exhibited a remarkable phagocytic effect against retinoblastoma Y79 cells and retinoblastoma organoids, as demonstrated by live imaging. Moreover, GD2 CAR-microglia administration to immunodeficient mice carrying a retinoblastoma xenograft resulted in a significant reduction in tumor growth and prolonged survival.

CONCLUSIONS. Our findings demonstrate a potent antitumor effect of GD2-CAR–engineered microglia in retinoblastoma, offering compelling evidence to support the continued development of CAR-microglia as a therapeutic strategy for this disease.

Keywords: retinoblastoma, chimeric antigen receptor, microglia, induced pluripotent stem cells

Retinoblastoma is primarily an intraocular malignancy in children that is fatal if left untreated.¹ Current treatments can be severe, including enucleation, which significantly impacts patients' quality of life.² Despite recent advances in conventional therapies, such as surgery, radiotherapy, and chemotherapy, the prognosis for patients with these tumors remains poor.³

As chimeric antigen receptor (CAR)–T cell therapy opened a new era of cancer immunotherapy for hematologic malignancies,⁴ multiple CARs have been developed to target tumors,^{5,6} including improving T-cell capacity using immune checkpoint inhibitors, using T-cell subpopulations and NK cells, or making off-the-shelf universal cells.^{7,8} CAR-expressing T cells and NK cells represent a new family of engineered immune cells with potent cytotoxic activity against cancer cells.^{9–11} However, their application in solid tumors has been challenging due to a variety of barriers, including limited tumor penetration, an immunosuppressive microenvironment, antigen escape, and tumor heterogeneity.^{12–14}

The high proportion of macrophages in tumor tissues indicates that macrophages are readily able to enter solid tumors.¹⁵ Tumor-associated macrophages are associated

with tumor microenvironment (TME) inflammation, particularly by increasing cytokines and promoting immune escape.^{16,17} CAR-macrophages have been utilized as effector immune cells to eliminate cancer cells by trafficking to the tumor site, then phagocytosing and activating the tumor microenvironment.⁹

Microglia, the resident macrophages in the retina, play a crucial role in maintaining tissue homeostasis and responding to pathological states.^{18–20} Their dysfunction can lead to pathological inflammation, which significantly impacts disease progression.^{21–23} Derived from the yolk sac, microglia express unique markers such as TMEM119 and HEXB and coexpress M1 (proinflammatory) and M2 (anti-inflammatory) markers, which are different from peripheral macrophages.^{20,24} Human microglia are difficult to obtain and culture, making microglia derived from human induced pluripotent stem cells (iPSCs) an effective research tool.^{25–27} However, whether microglia can be effectively engineered to target and eliminate retinoblastoma remains unexplored.

In this study, we develop CAR-microglia using ganglioside GD2 as the antigen due to its high and selective expression in retinoblastoma.²⁸ By manufacturing GD2 CAR-microglia using a human microglia cell line and iPSC-derived microglia

(iMGs), we reveal the antigen-dependent antitumor effects of GD2 CAR-microglia in cocultured retinoblastoma cells Y79, cocultured retinoblastoma organoids, and a retinoblastoma xenograft model. Our findings provide a novel and effective treatment strategy for retinoblastoma, which may pave the way for the development of CAR-microglia-based therapies as a promising approach for patients with currently untreatable retinoblastoma.

METHODS

Cell Culture

This study was approved by the institutional ethical committee and conducted in accordance with the Declaration of Helsinki. The human microglial cell line HMC3 was purchased from Procell (Wuhan, China). Y79 and HEK293T cells were bought from ATCC (Manassas, VA, USA). HMC3, Y79, and HEK293T cells were maintained in MEM, RPMI-1640, and Dulbecco's modified Eagle's medium, respectively, all supplemented with 10% to 20% FBS and 1% penicillin-streptomycin (Gibco, Grand Island, NY, USA). The methods for maintaining human iPSCs were previously described.^{27,29}

Construction of CARs

We designed CARs according to the intracellular CD86 domain (amino acids 269–329) and FcγR domain (amino acids 314–374), integrating hinge and transmembrane regions of CD8α (amino acids 183–206) as previously described.¹⁰ The extracellular fragment of CARs comprised the humanized scFv nucleotide sequences specific to GD2 and CD19, as previously described.^{28,30}

Stable Overexpression of CARs in HMC3 Cells and iPSCs

HEK293T cells were cultured for CAR-lentivirus production. The concentration of lentivirus was 1×10^8 TU/mL. The 30 μL resuspended lentivirus was then added to prepared HMC3 cells or iPSCs in 6-well plates with polybrene. The medium was refreshed after 6 hours postinfection. The infected cells were cultured for 24 hours and then treated with 2 μg/mL or 0.5 μg/mL puromycin for at least 48 hours to select successfully transduced clones.

CAR-iMG Differentiation From CAR-iPSCs

The differentiation method of iPSCs into microglia was previously described.²⁷ Briefly, when CAR-engineered iPSCs reached 80% confluence, cells were detached using 0.5 mM EDTA (Invitrogen) and transferred to a plate with an iPS base medium. On day 5, cells were transferred into 0.1% gelatin-precoated (Sigma-Aldrich, St. Louis, MO, USA) tissue culture dishes to generate myeloid precursor cells in Lonza (Basel, Switzerland) x-vivo 15 plus medium supplemented with 0.055 mM β-mercaptoethanol, 2 mM L-glutamine, 1% penicillin-streptomycin, 25 ng/mL hIL-3 (R&D Systems, Minneapolis, MN, USA), and 50 ng/mL hM-CSF (R&D Systems). The medium was refreshed weekly. Hematopoietic progenitors emerged in the supernatant within 4 weeks. For CAR-iMG differentiation and maturation, the suspension cells were plated with microglial maturation medium. The medium was changed weekly, and CAR-iMGs were typically used for functional experiments after 1 week of incubation.

Antitumor Effect of CAR-iMGs

To assess the antitumor effect of CAR-iMGs, we cocultured CAR-iMGs with luc⁺ Y79 at the indicated effector-to-target (E/T) ratio in a 96-well plate in vitro and then added 100 μL D-luciferin sodium (MCE, HY-12591) into the culturing wells and captured the BLI signal from tumor cells (Molecular Devices, San Jose, CA, USA).

Antitumor assay

To evaluate the antitumor effect of GD2-targeting CAR-iMGs on Y79 cells, we cocultured enhanced green fluorescent protein positive (EGFP⁺) CAR-iMGs with CellTrace Violet-stained Y79 cells at an E/T ratio of 1/1 for flow cytometry, live imaging, and RNA sequencing (RNA-seq).

Retinoblastoma Organoid Differentiation

The retinoblastoma organoid differentiation method has been described previously.^{29,31} Briefly, *RBI*^{-/-} stem cells were centrifuged and cell pellets were resuspended in ice-cold Matrigel, polymerized at 37°C for 20 minutes, and then floated in N2B27 medium (day 1). On day 5, the medium was refreshed. From day 8, the cysts adhered to the bottom of the dish and began to grow. On day 15, the adherent cells were digested with dispase, then grown in B27 medium. Medium was changed weekly using serum medium. Tumors appeared in the retinoblastoma organoids after differentiation for about 60 days.

In Vivo Antitumor Assays

C-NKG immunodeficient mice were bought from Cyagen Biosciences (Suzhou, China) and maintained under pathogen-free conditions at the animal center of the Beijing Tongren Hospital. Y79 luc⁺ cells (1×10^4 cells) were engrafted into the subretinal space of female C-NKG immunodeficient mice (6 weeks of age). Two weeks later, mice were injected with either GD2 or CD19 CAR-iMGs (1×10^6 cells) into the vitreous. Mice received 150 mg/kg D-luciferin intraperitoneally, and BLI was acquired on a Vilber Fusion FX SPECTRA imager (Vilber, Paris, France). An optical coherence tomography (OCT) image (Leica envisu R2210UHR; Leica, Wetzlar, Germany) and fundus photograph (Micron III; Phoenix MICRON, Oregon, USA) were obtained at the indicated time point.

Immunofluorescence Assay and Confocal Microscopy Imaging

The indicated cells were attached to glass coverslips in 24-well plates for the immunofluorescence assay. The method of the immunofluorescence assay and confocal microscopy imaging was previously described.²⁷ Fluorescence was analyzed using an Olympus (Tokyo, Japan) confocal microscope.

RNA Sequencing and Analysis

The indicated cells were isolated using TRIzol reagent (Invitrogen, Carlsbad, CA, USA). RNA purity and concentration were then examined using NanoDrop 2000. Library construction and sequencing were performed on the Illumina platform, and gene fragments per kilobase of transcript per million mapped reads (FPKM) were calculated.

from read counts mapped to each gene and the gene length (Berry Genomics, Beijing, China). The significantly differential expression was assigned at an adjusted $P < 0.05$.

Quantitative Real-Time PCR Assay

The method of quantitative real-time PCR (qRT-PCR) analysis was previously described.²⁷ The following primers were used for human *GAPDH*, *MERTK*, *AXL*, *CD68*, *CASP8*, *FAS*, *GSDMD*, *RELA*, *NFKB2*, *IKBK*, *IL12A*, *IL6*, *MAPK3*, *MAPK7*, *MAPK8*, and *MAPK12*:

Gene	Primer Forward 5'-3'	Primer Reverse 5'-3'
<i>GAPDH</i>	GGAGCGAGATCCCTCCAAAAT	GGCTGTTGTCATACTTCTCATGG
<i>MERTK</i>	AGCCATTGAACCTTACCTTA	TCCTTCCATTACAGACCC
<i>AXL</i>	CTCCTGCGAAGCCCATAA	GACCAGGTGGAGGTTACGG
<i>CD68</i>	ACGCAGCACAGTGACAT	AGGCCAAGAAGGATCAGG
<i>CASP8</i>	GGGGTAATGACAATCTCG	CTCTTCAAAGGTCGTGGT
<i>FAS</i>	TGGCTTAGAAGTGAAAT	CTTGGTGTGCTGGTGAG
<i>GSDMD</i>	GACAACGTGTAGTGGTGACTG	TGGAAGTCTCTGCTTCTTAT
<i>RELA</i>	GGGGACTACGACCTGAATG	TGCCAGAGTTTCGGTTCA
<i>NFKB2</i>	CGATTTCGATATGGCTGTG	GGCTGGTCCCTCGTAGTT
<i>IKBK</i>	CACCAGCTCTTCCAAGAATACG	CGGAACGGTCTCCATCACA
<i>IL12A</i>	CTCCTTGTGGCTACCTG	AGTTGTCTGGCCTTCTG
<i>IL6</i>	CAAATTCGGTACATCCTC	CTGGCTTGTCTCTCACTA
<i>MAPK3</i>	AACACCACCTGCGACCTT	CGTAGCCACATACTCCGTCA
<i>MAPK7</i>	GTGGGCGACGAGTACGAGA	GAGGGTCCGCTTGGCAAT
<i>MAPK8</i>	ACACCACAGAAATCCCTA	CTTTAAGTCCCGATGAATA
<i>MAPK12</i>	GCAAGACGCTGTTCAGGG	TGGTCAGGATAGAGGCAAAA

ELISA

The concentrations of human IL-6 and TNF- α in the indicated cell culture supernatants were quantified using ELISA kits (BioLegend, San Diego, CA, USA) according to the manufacturer's instructions.

Flow Cytometry Analysis

The indicated cells were prepared according to the manufacturer's instructions. Then the samples were analyzed on an Attune NxT flow cytometer (Invitrogen), and the resulting data were analyzed using the Attune Nxt software (Invitrogen).

Statistical Analysis

The two-tailed Student's *t*-test was used to analyze statistical significance between two groups. ANOVA was used to analyze statistical significance in three groups. $P < 0.05$ (*), $P < 0.01$ (**), and $P < 0.001$ (***) were considered statistically significant.

RESULTS

CAR-Microglia Were Successfully Generated From Microglia Cells

GD2 was selected as the retinoblastoma antigen due to its selective expression in a wide range of central nervous system (CNS) tumors, including retinoblastoma.³² CD19 was used as the control antigen since it is not expressed in retina, while CD19 CAR T-cell therapy in B-cell lymphomas has been well recognized.³³ The expression profile of GD2 and CD19 in retinoblastoma (Y79) and microglia cells (HMC3, iMGs) is shown in Supplementary Figure S1.

First, we used microglial HMC3 cells for screening antitumor functions because they offer advantages in cell culture and gene transduction compared to primary microglial cells. We generated lentivirus containing a sequence encoding GD2 scfv-specific CAR^{10,30} and CD19 scfv-specific CAR²⁸ with EGFP (Fig. 1A). The expression of CAR in HMC3 cells was verified using a fluorescence microscope (Fig. 1B) and PCR (Fig. 1C). Immunofluorescence exhibited the expression of CAR-EGFP and microglia markers, including IBA1 and CX3CR1 (Fig. 1D). In addition, flow cytometry analysis demonstrated the expression of CAR-EGFP and CAR-specific linker G4S (Figs. 1E, 1F). These data demonstrated that we generated GD2 CAR-microglia and CD19 CAR-microglia.

GD2 CAR-Microglia Exerted Phagocytic Function Against Retinoblastoma Cells

To investigate the antitumor efficacy of CAR-microglia, we cocultured Y79 cells (stained with CellTrace Violet) with GD2 CAR-microglia and control CD19 CAR-microglia (Fig. 2A). GD2 CAR-microglia phagocytosed Y79 cells significantly (Fig. 2B and Supplementary Movie S1) but not CD19 CAR-microglia (Fig. 2B and Supplementary Movie S2) within 6 hours. Similar results were discovered by flow cytometry analysis. Specifically, we observed more EGFP⁺ CellTrace Violet⁺ cells, which represent CellTrace Violet⁺ Y79 cells engulfed by EGFP⁺ CAR-microglia in the cocultured GD2 CAR-microglia compared with CD19 CAR-microglia at 24 hours (Figs. 2C, 2D). In addition, the number of Y79 cells was significantly decreased when coculturing with GD2 CAR-microglia compared to CD19 CAR-microglia by 48 hours (Figs. 2E, 2F).

Moreover, the antigen-dependent anti-retinoblastoma ability of CAR-microglia was further confirmed by coculture with Y79 at different E/T ratios (Figs. 2G, 2H). However, IL-6 and TNF- α concentration in the culture medium did not differ between the GD2 CAR and CD19 CAR-microglia coculture (Fig. 2I), suggesting that the antitumor effect of GD2 CAR-microglia is mainly due to phagocytosis. RNA-seq analysis revealed upregulation of cell death-related genes (*CASP3*, *GSDMD*, *IL1A*) and microglia activation-related MAPK (*MAPK1*, *MAPK7*, *MAPK10*, *MAPK11*, *MAPK14*) and NF- κ B (*NFKB1*, *TRAF3*, *TRAF6*, *TRAF3IP1*) signaling pathway genes in cocultured GD2 CAR-microglia compared to CD19 CAR-microglia (Supplementary Fig. S2). These data demonstrated that GD2 CAR-microglia had antitumor cell functions mainly due to phagocytosis.

CAR-iMGs Were Successfully Generated From iPSCs

As immortalized microglia cell lines cannot be applied to clinical settings and primary microglia are hard to obtain, iMGs are a great source for microglial-based immunotherapy. Human iPSCs were infected with lentivirus carrying the sequence that encodes either GD2 scfv-specific CAR or CD19 scfv-specific CAR, then differentiated to microglia as previously reported (Fig. 3A).^{26,27} The expression of microglia phenotypes IBA1 and CX3CR1 was verified in GD2 CAR-iMGs and CD19 CAR-iMGs by immunofluorescence (Fig. 3B). Flow cytometry verified the microglia purity (CD45⁺CD11b⁺) in GD2 CAR-iMGs and CD19 CAR-iMGs

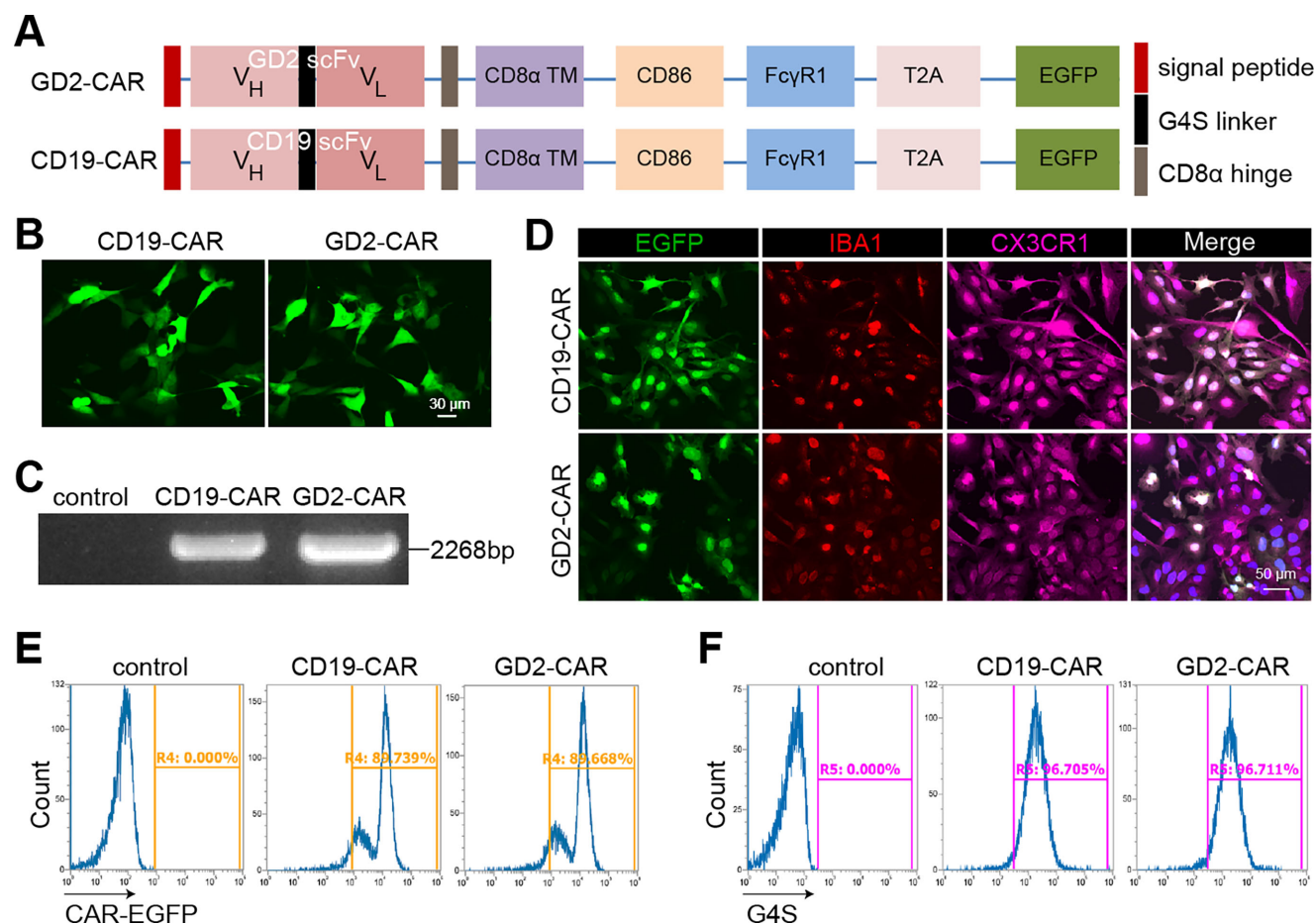


FIGURE 1. Manufacturing of GD2 CAR-microglia using a microglia cell line. (A) Constructs of GD2/CD19-targeting CARs. The CARs mainly comprise an extracellular signal peptide (red rectangle), a scFv recognizing GD2 or CD19, a transmembrane (TM) domain and a hinge region from CD8α, and a cytoplasmic portion of CD86 plus the cytoplasmic part of the human FcγRI. The G4S linker was between V_H and V_L to form scFv. The CD8α hinge was between scFv and CD8α TM domains to endow the CARs with the flexibility to target antigens. These CAR transgenes were linked to the EGFP-expressing sequence via a T2A element. (B) Representative images of microglia cell line HMC3 infected with GD2-CAR lentivirus or CD19-CAR lentivirus. (C) Representative images of the PCR product of GD2-CAR or CD19-CAR in HMC3 cells infected with GD2-CAR lentivirus or CD19-CAR lentivirus. (D) Representative images of GD2-CAR HMC3 or CD19-CAR HMC3 cells immunostained for IBA1 (red) and CX3CR1 (rose red). Cell nuclei were stained with DAPI (blue). Scale bar: 50 μm. (E, F) Fluorescence-activated cell sorting (FACS) analysis of CAR-EGFP (E) and linker G4S (F) expression in GD2-CAR HMC3 or CD19-CAR HMC3 cells.

(Figs. 3C, 3D). The transduction efficiency of the CAR construct (EGFP⁺) was more than 85% in GD2 CAR-iMGs and CD19 CAR-iMGs (Figs. 3E, 3F). In addition, we also performed RNA-seq to analyze gene expression in different steps during the differentiation of CAR-iMGs derived from iPSCs. RNA-seq analysis showed specific expression of microglia marker genes (*P2RY12*, *AIF1*, *TMEM119*, *HEXB*) in GD2 CAR-iMGs and CD19 CAR-iMGs (Supplementary Fig. S3A). We also found that the expression of microglia activation-related MAPK (*MAPK3*, *MAPK13*, *MAPK14*) and NF-κB (*NFKB1*, *TRAF3*, *RELA*) signaling pathway genes was comparable in control iMGs and GD2 or CD19 CAR-iMGs (Supplementary Fig. S3B). Furthermore, the CAR-iMG transcriptome was profiled in comparison to human fetal microglia and adult microglia from previous studies (GSE89189).³⁴ The correlation analysis showed that our CAR-iMGs were highly correlated with human microglia (Supplementary Fig. S3C). These data demonstrated that we successfully generated GD2 CAR-iMGs and CD19 CAR-iMGs.

GD2 CAR-iMGs Exerted Phagocytic Function Against Y79 Cells

To investigate the antitumor efficacy of CAR-iMGs, we cocultured Y79 cells with GD2 CAR-iMGs and control CD19 CAR-iMGs (Fig. 4A). We then detected the EGFP⁺ CAR-iMG phagocytic CellTrace Violet⁺ Y79 cells (EGFP⁺ CellTrace Violet⁺) using fluorescence-activated cell sorting and confocal microscope analysis. We discovered that significantly increased Y79 were phagocytosed when coculturing with GD2 CAR-iMGs compared to CD19 CAR-iMGs for 24 hours (Figs. 4B–D). We also found that the microglial phagocytosis-related genes *MERTK*, *AXL*, and *CD68* were upregulated in GD2 CAR-iMGs cocultured with Y79 compared to CD19 CAR-iMGs for 24 hours by qRT-PCR (Fig. 4E). Moreover, the antigen-dependent anti-retinoblastoma ability of CAR-iMGs was further confirmed by coculture with Y79 at different E/T ratios (Figs. 4F, 4G). Then we performed RNA-seq to analyze the microglial function in CAR-iMGs cocultured with or without Y79 cells. We found that cell death-related

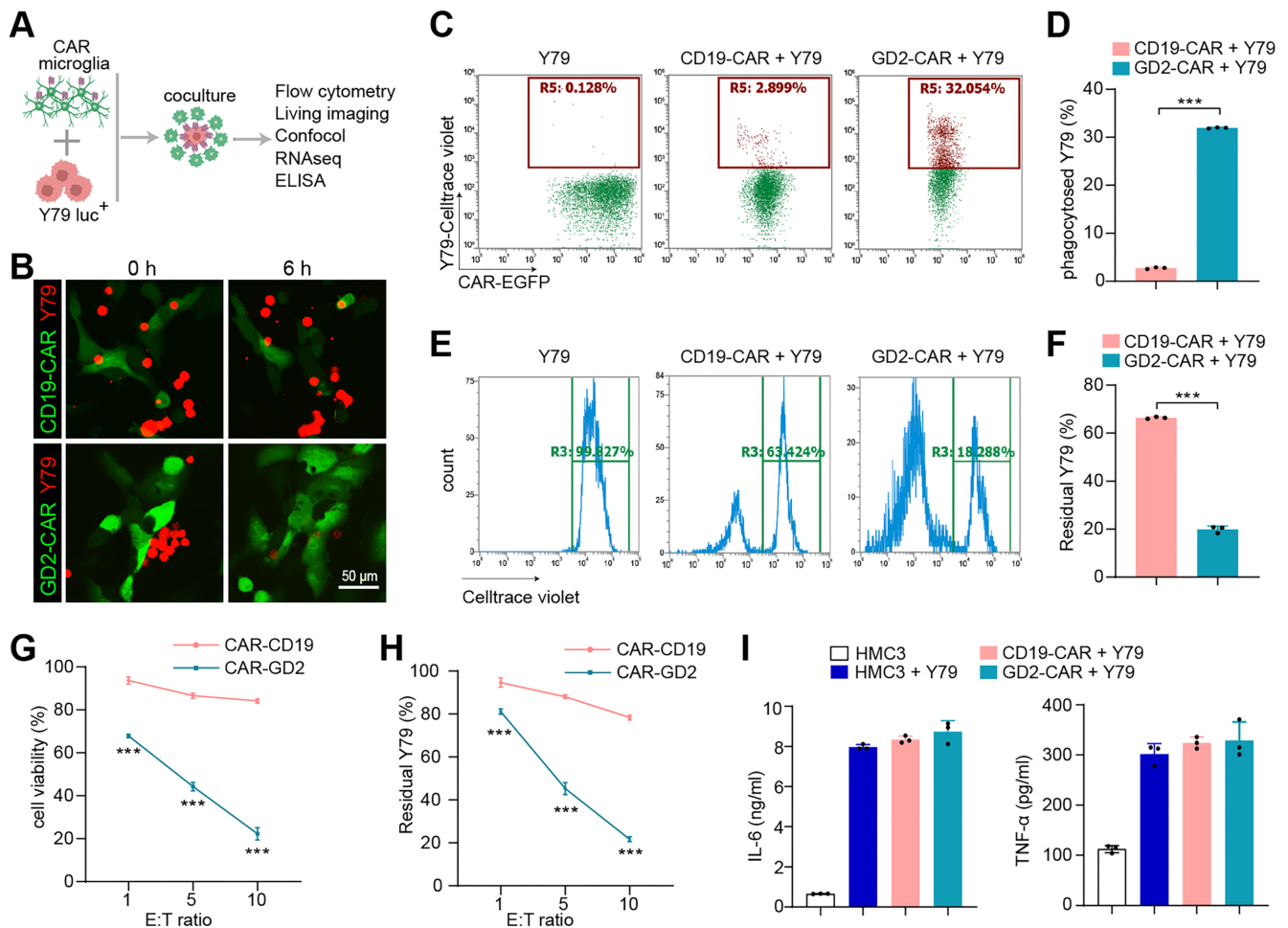


FIGURE 2. Effects of GD2 CAR-microglia derived from a microglia cell line in coculture. (A) Schematic diagrams of CAR-microglia and tumor cells in coculture experiments. (B) Representative images of GD2 CAR-microglia (green, namely GD2-CAR) or CD19 CAR-microglia (green, namely CD19-CAR) cocultured with a retinoblastoma cell line Y79 (red) stained with CellTrace Violet for 0 and 6 hours by confocal live imaging. (C) FACS analysis of Y79 cells phagocytosed by CAR-microglia after coculturing with EGFP⁺ GD2 CAR-microglia or EGFP⁺ CD19 CAR-microglia for 24 hours. R5 gate represents Y79 cells phagocytosed by CAR-microglia (%). (D) Quantification of Y79 cells phagocytosed by CAR-microglia after coculturing with EGFP⁺ GD2 CAR-microglia or EGFP⁺ CD19 CAR-microglia for 24 hours. (E) FACS analysis of residual Y79 cells after coculturing with GD2 CAR-microglia or CD19 CAR-microglia. R3 gate represents residual Y79 cells (%). (F) Quantification of the residual Y79 cells after coculturing with or without GD2 CAR-microglia or CD19 CAR-microglia. (G) Quantification of the Y79 luc⁺ cell viability after coculturing with GD2 CAR-microglia or CD19 CAR-microglia for 48 hours by CCK8 analysis. (H) Quantification of the residual Y79 luc⁺ cells after coculturing with GD2 CAR-microglia or CD19 CAR-microglia for 48 hours by bioluminescence analysis. (I) ELISA analysis of IL-6 (left) and TNF-α (right) protein levels in the supernatant of GD2 CAR-microglia or CD19 CAR-microglia with or without Y79 for 48 hours; HMC3 and HMC3 cocultured with Y79 were used as a control. Data are shown as mean ± SD (*n* = 3). ***P* < 0.01, ****P* < 0.001.

genes (*GSDMD*, *FAS*, *CASP8*, *IL1A*) and microglia activation-related MAPK (*MAPK3*, *MAPK7*, *MAPK8*, *MAPK12*) and NF-κB (*NFKB2*, *RELA*, *IKBK*) signaling pathway genes were upregulated in GD2 CAR-iMGs cocultured with Y79 cells (Fig. 4H). To confirm the RNA-seq results, we performed qRT-PCR to verify the expression of *CASP8*, *FAS*, *GSDMD*, *RELA*, *NFKB2*, *IKBK*, *IL12A*, *IL6*, *MAPK3*, *MAPK7*, *MAPK8*, and *MAPK12*. As shown in Supplementary Figure S4, the mRNA expression of *CASP8*, *FAS*, *GSDMD*, *RELA*, *NFKB2*, *IL12A*, *MAPK3*, *MAPK7*, *MAPK8*, and *MAPK12* was significantly upregulated in GD2 CAR-iMGs cocultured with Y79 cells. Interestingly, IL-6 expression was reduced in GD2 CAR-iMGs and Y79 cells cocultured with supernatant compared with the CD19 CAR-iMG cocultured group, which is distinct from the previous results (Fig. 4I).

GD2 CAR-iMGs Exerted Phagocytic Function Against Retinoblastoma Organoids

Next, we examined the effects of GD2 CAR-iMGs using human retinoblastoma organoids (hRBOs), which were derived from *RBI*^{-/-} stem cells, as previously reported (Fig. 5A).^{29,31} The expression of GD2 instead of CD19 was detected in hRBOs (Fig. 5B). Then we cocultured hRBOs with GD2 CAR-iMGs or CD19 CAR-iMGs for 10 days. CD19 CAR-iMGs promoted the formation of hRBOs (Figs. 5C, 5D). In contrast, GD2 CAR-iMGs disrupted tumor formation and reduced tumor growth (Figs. 5C, 5D). The live imaging system also showed promotion of tumor organoid formation upon coculturing with CD19 CAR-iMGs (Figs. 5E, 5F; Supplementary Movie S3). However, hRBOs were significantly inhibited in GD2 CAR-iMG treatment (Figs. 5E, 5F; Supple-

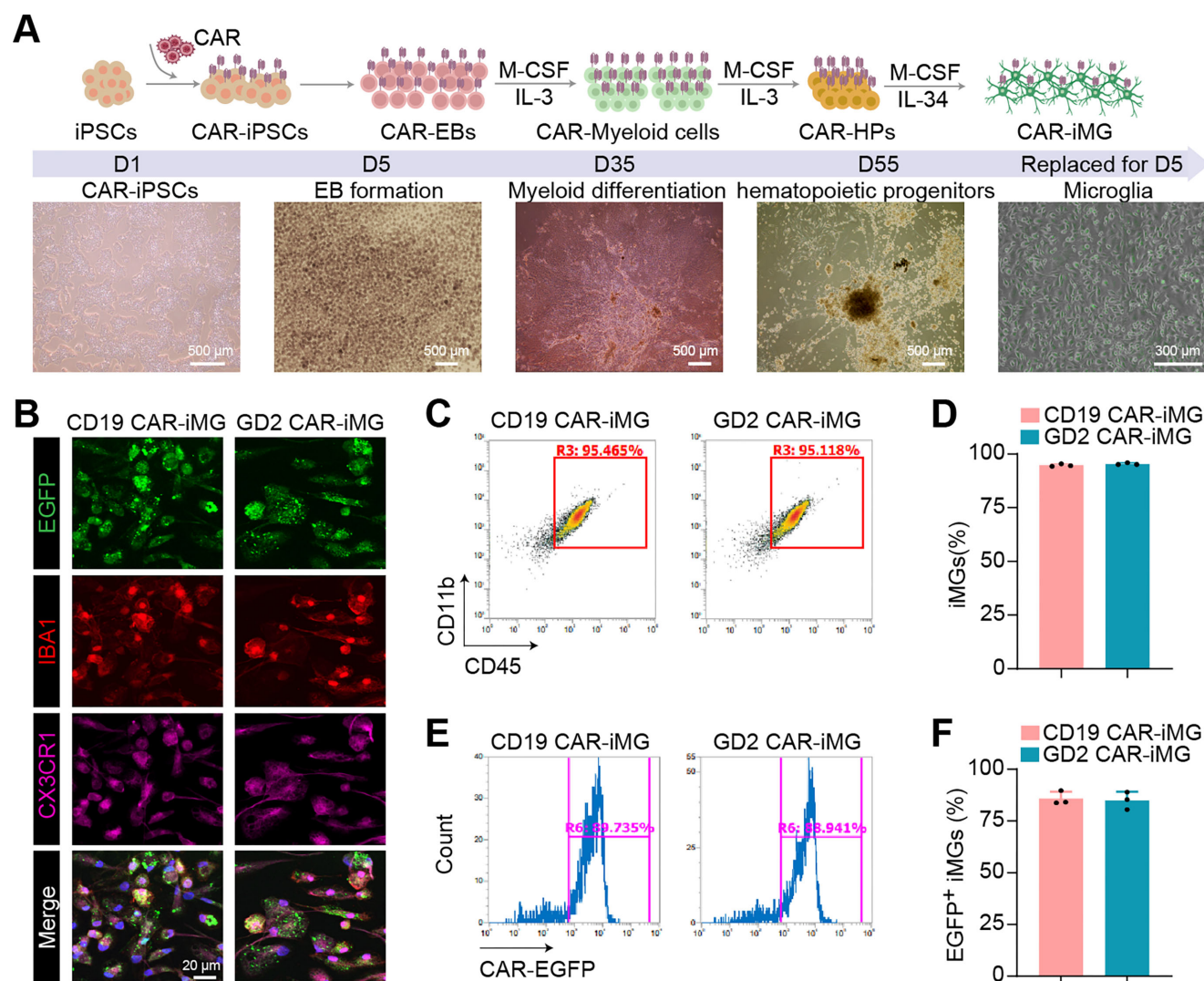


FIGURE 3. Manufacturing of GD2 CAR-iMGs using iPSCs. (A) Overview of the differentiation process of CAR-iMGs from the CAR-expressing iPSCs. EGFP-labeled CAR-iPSCs were differentiated into CAR-iMGs by mimicking the in vivo process of embryoid bodies (EBs), myeloid differentiation, hematopoietic progenitors, and microglia-like cells. (B) Representative images of GD2 CAR-iMGs or CD19 CAR-iMGs immunostained for microglia markers IBA1 (red) and CX3CR1 (rose red). Cell nuclei were stained with DAPI (blue). Scale bar: 50 μ m. (C) FACS analysis of the purity of iMGs derived from GD2 or CD19 CAR-expressing iPSCs using CD45 and CD11b markers. (D) Quantification of the GD2 or CD19 CAR-expressing iMGs. (E) FACS analysis of CAR-EGFP transduction efficiency in GD2 CAR-iMGs or CD19 CAR-iMGs. (F) Quantification of EGFP⁺ CAR-iMGs in GD2 CAR-iMGs or CD19 CAR-iMGs.

mentary Movie S4). IL-6 concentration in the supernatant was significantly lower in hRBOs cocultured with GD2 CAR-iMGs than CD19 CAR-iMGs (Supplementary Fig. S5A). RNA-seq analysis showed upregulation of microglia activation-related MAPK (*MAPK3*, *MAPK7*, *MAPK8*, *MAPK10*, *MAPK11*, *MAPK12*) and NF- κ B (*NFKB2*, *IKBK*, *RELA*) signaling pathway genes in CAR-iMGs cocultured with hRBOs compared with CAR-iMGs (Supplementary Fig. S5B).

Efficacy of GD2 CAR-iMG Therapy in the Retinoblastoma Xenograft Model

To investigate the antitumor effect of GD2 CAR-iMGs in vivo, we established a Y79 retinoblastoma xenograft model. Two weeks later, mice were intravitreally injected with either GD2 or CD19 CAR-iMGs (Fig. 6A). The therapeutic effi-

cacy was comprehensively evaluated using a combination of advanced imaging techniques and histologic analysis. OCT provided high-resolution cross-sectional images of the retina. We observed a significant tumor reduction in the GD2 CAR-iMG-treated group after 14 days, indicating a marked inhibition of tumor proliferation compared to the control group (Fig. 6B). Fundus photography further revealed a diminished tumor size and altered vascular architecture in the treated mice, suggesting a robust antitumor response (Fig. 6C). Through in vivo bioluminescence imaging, we observed a significant reduction in bioluminescent signals in the GD2 CAR-iMG-treated group, confirming the reduction in tumor volume (Figs. 6D, 6E). Moreover, the GD2 CAR-iMG-administrated mice showed prolonged survival, compared to the CD19 CAR-iMG treatment group (Fig. 6F). Histologic examination using hematoxylin and eosin staining further corroborated these findings (Fig. 6G).

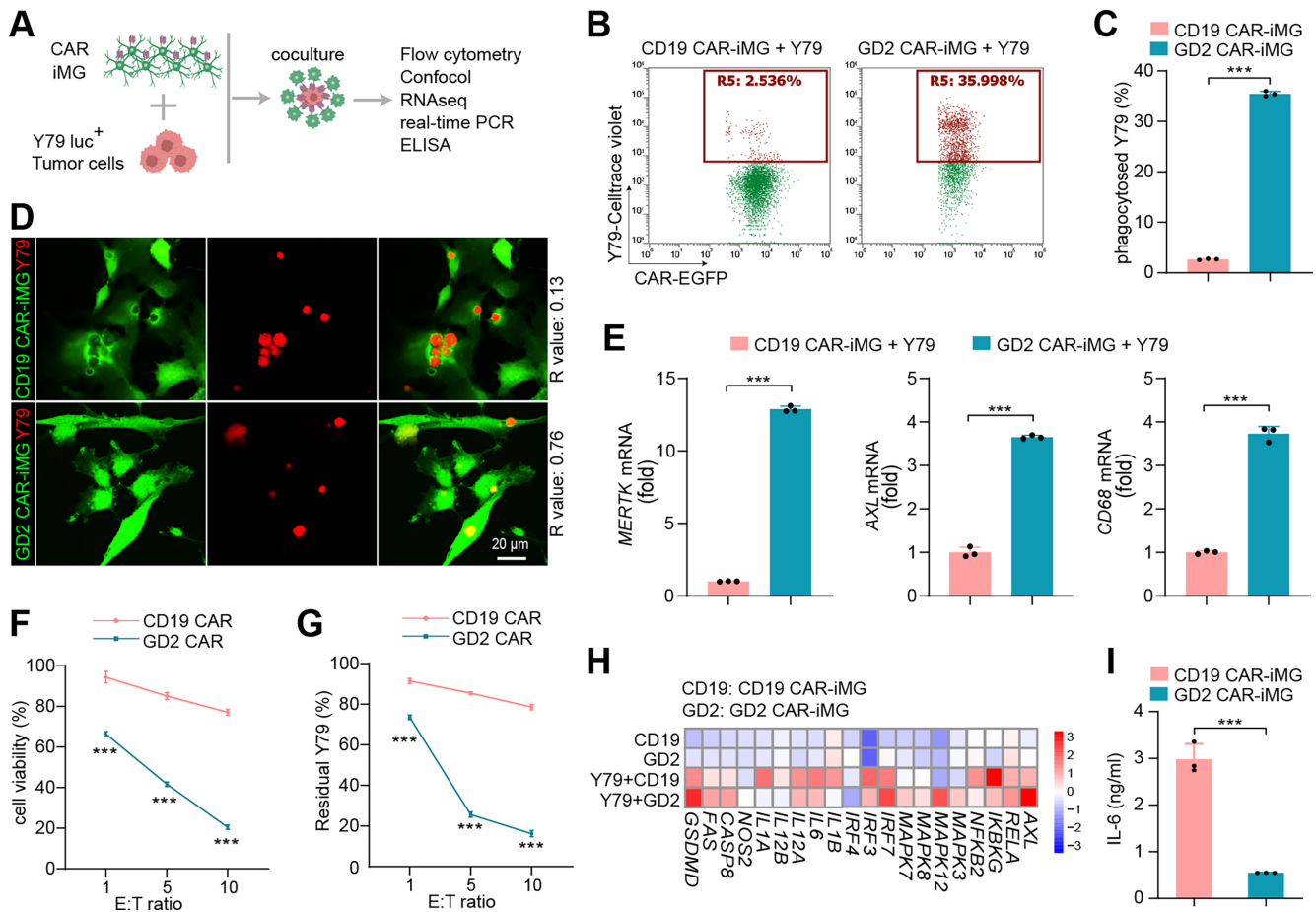


FIGURE 4. Effects of GD2 CAR-microglia derived from iPSCs in coculture. (A) Schematic diagrams of CAR-iMGs and tumor cell coculture experiments. (B) FACS analysis of Y79 cells phagocytosed by CAR-iMGs after coculturing with EGFP⁺ GD2 CAR-iMGs or EGFP⁺ CD19 CAR-iMGs for 24 hours. (C) Quantification of Y79 cells phagocytosed by CAR-iMGs after coculturing with EGFP⁺ GD2 CAR-microglia or EGFP⁺ CD19 CAR-microglia for 24 hours. (D) Representative confocal microscopy analysis images of GD2 CAR-iMGs (green) or CD19 CAR-iMGs (green) cocultured with Y79 cells (red) stained with CellTrace Violet dye for 24 hours. The *R* value represents the Pearson's correlation coefficient of the Y79 cell signal within CAR-microglia. (E) The qRT-PCR assay of *MERTK*, *AXL*, and *CD68* expression in GD2 CAR-iMGs or CD19 CAR-iMGs cocultured with Y79 cells for 24 hours. (F) Quantification of the Y79 luc⁺ cell viability after coculturing with GD2 CAR-iMGs or CD19 CAR-iMGs for 48 hours by CCK8 analysis. (G) Quantification of the residual Y79 luc⁺ cells after coculturing with GD2 CAR-iMGs or CD19 CAR-iMGs for 48 hours by bioluminescence analysis. (H) Heatmap analysis of microglia activation-related cytokines, MAPK, and NF- κ B signaling pathway-related gene expression in tumor cells cocultured with CAR-iMGs. (I) ELISA analysis of IL-6 protein levels in the supernatant of CAR-CD19 iMG or CAR-GD2 iMG cells with Y79 (h) for 48 hours. Data are shown as mean $\pm$ SD ($n = 3$). *** $P < 0.01$, **** $P < 0.001$.

DISCUSSION

Genetically engineered CAR macrophages have drawn substantial attention in solid tumors,^{9,35} which show potential advantages in homing and infiltrating within solid tumors and manifest roles in reversing the immunosuppressive TME. The antitumor effects of CAR-macrophages have been reported in ovarian/pancreatic cancer cells¹⁰ and hepatocellular carcinoma.¹¹ Microglia, as the primary resident immune cells in neuronal tissues, play critical roles in CNS homeostasis.^{36–38} It has been reported recently that human iPSC-derived microglia can be engineered to enable pathology-responsive and CNS-wide delivery of therapeutic proteins.³⁹ Their findings suggested that iMGs could offer a promising new platform technology to deliver biological therapeutics throughout the brain.

In our study, we found engineered CAR-microglia showed antitumor effects in cocultured Y79, retinoblastoma organoids, and a retinoblastoma xenograft model, there-

fore providing convincing results for the potential application in brain tumor treatment. GD2 has been reported as the specific marker for multiple tumors, as well as the potential target for antitumor therapy.⁴⁰ There are many methods to differentiate microglia from iPSCs, and the differentiation time is long and the purity yield is low.²⁵ The microglia differentiation system needs to be optimized to shorten the differentiation time and improve the yield. One side effect of CAR-T treatment is cytokine release syndrome, which can lead to multiple organ failure if not treated promptly.⁴¹ Our results suggested that localized delivery of GD2 CAR-iMGs has a more powerful antitumor ability and lower cytokine expression than CD19 CAR-iMGs. In future studies, it would be important to compare the safety of GD2-CAR T cells and GD2-CAR microglia.

Interestingly, unlike coculturing with tumor cell lines, we found that tumor proliferation was accelerated in CD19 CAR-iMGs cocultured with hRBOs, indicating the protumor effect

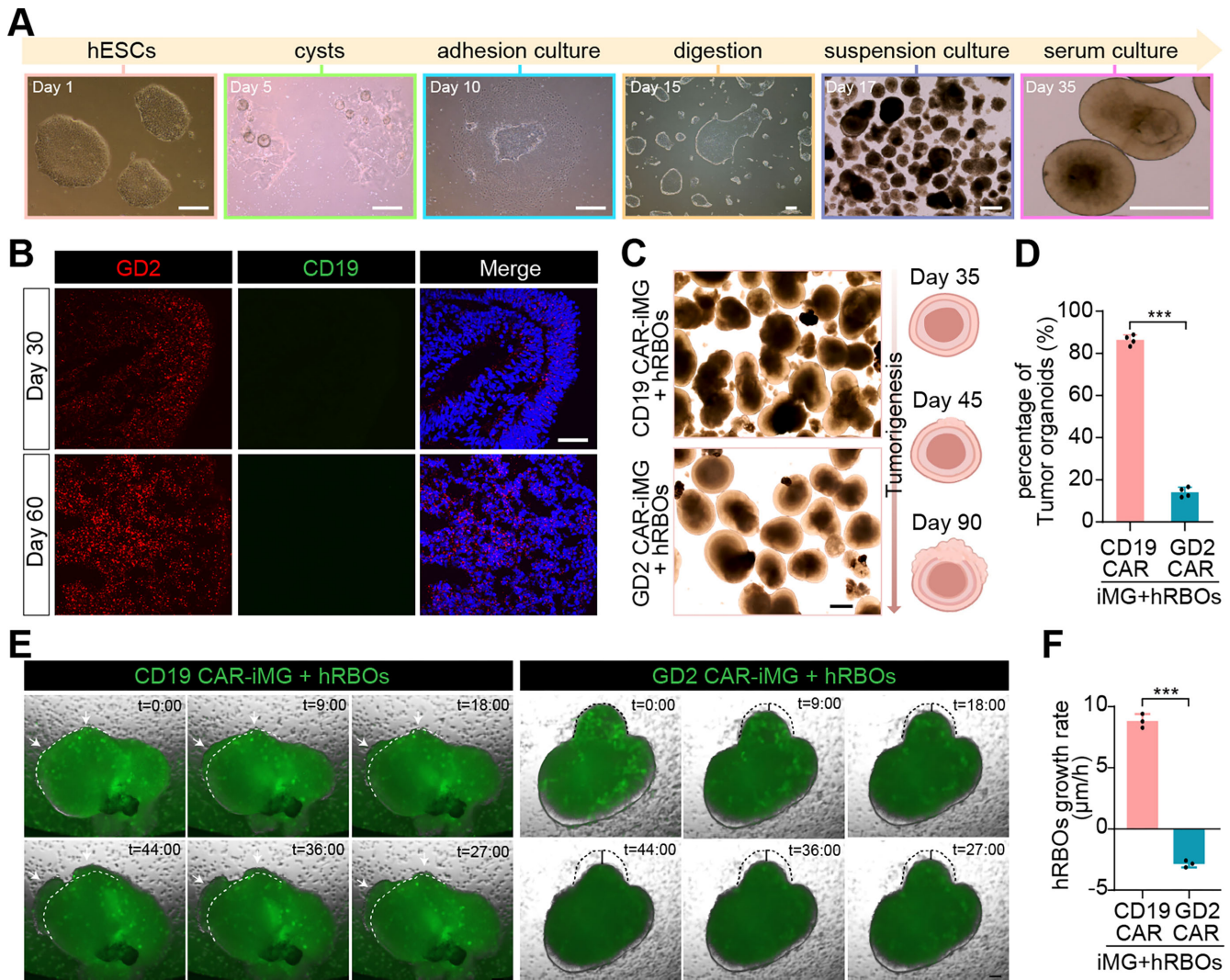


FIGURE 5. Effects of GD2 CAR-microglia derived from iPSCs in organoids. (A) Schematic of the stages and representative images along the timeline during differentiation of human embryonic stem cells (hESCs) into hROs. Scale bar: 200 μ m. (B) Representative images of hRBOs immunostained for GD2 (red) and CD19 (green). Scale bar: 50 μ m. (C) Representative images of hRBOs cocultured with GD2 CAR-iMGs or CD19 CAR-iMGs for 10 days. Schematic diagrams on the right show the process of tumorigenesis in hRBOs. Scale bar: 500 μ m. (D) Quantification of the hRBOs after coculturing with GD2 CAR-iMGs or CD19 CAR-iMGs for 10 days. (E) Live imaging showing the time-dependent change of hRBOs (brightfield) cocultured with CD19 CAR-iMGs (green) or GD2 CAR-iMGs (green) in a BioTek Cytation C5 analysis instrument (Agilent, Santa Clara, CA, USA). White arrows and white dashed curves in hRBOs cocultured with CD19 CAR-iMGs indicate the initial tumor. The black dashed curves in hRBOs cocultured with GD2 CAR-iMGs indicate the initial tumor. The black straight line indicates the changes over time. Scale bar: 100 μ m. (F) Quantification of hRBO growth rate after coculturing with CD19 CAR-iMGs or GD2 CAR-iMGs. Data are shown as mean $\pm$ SD ($n = 3$). ** $P < 0.01$, *** $P < 0.001$.

of inappropriate treatment. The US Food and Drug Administration Modernization Act 2.0 has allowed the utilization of organoids for assessing drug safety and effectiveness.⁴² Our results confirmed that organoids can reflect the real situation more than cell lines.

While our study demonstrated that GD2-CAR-engineered microglia can suppress retinoblastoma growth, several limitations should be acknowledged. All functional assays were performed with Y79 cells or retinoblastoma organoids. Although Y79 remains the most widely characterized retinoblastoma line and our retinoblastoma organoids recapitulate key histologic features of the primary tumor, we cannot exclude the possibility that the observed sensitivity to GD2-CAR-microglia is unique to these genetic backgrounds. The absence of additional established lines (e.g.,

WERI-Rb1) or, more importantly, of freshly isolated patient-derived xenograft (PDX) cells limits the generalizability of our findings. Future work will expand the panel to multiple cell lines and integrate PDX-organoid platforms to confirm CAR-microglia activity.

In summary, our findings demonstrated that GD2 CAR-microglia confer antitumor activities in retinoblastoma, providing a potential strategy for immunotherapy of brain tumors. The persistence and efficiency of CAR-microglia need to be further improved by identifying more effective CARs or modifying the universal CAR-microglia in future studies. Systemic lupus erythematosus can be treated by CAR T cells.⁴³ Given that microglia activation is observed in various neurodegenerative diseases, can retinal degeneration diseases be treated by CAR-microglia? Identifying addi-

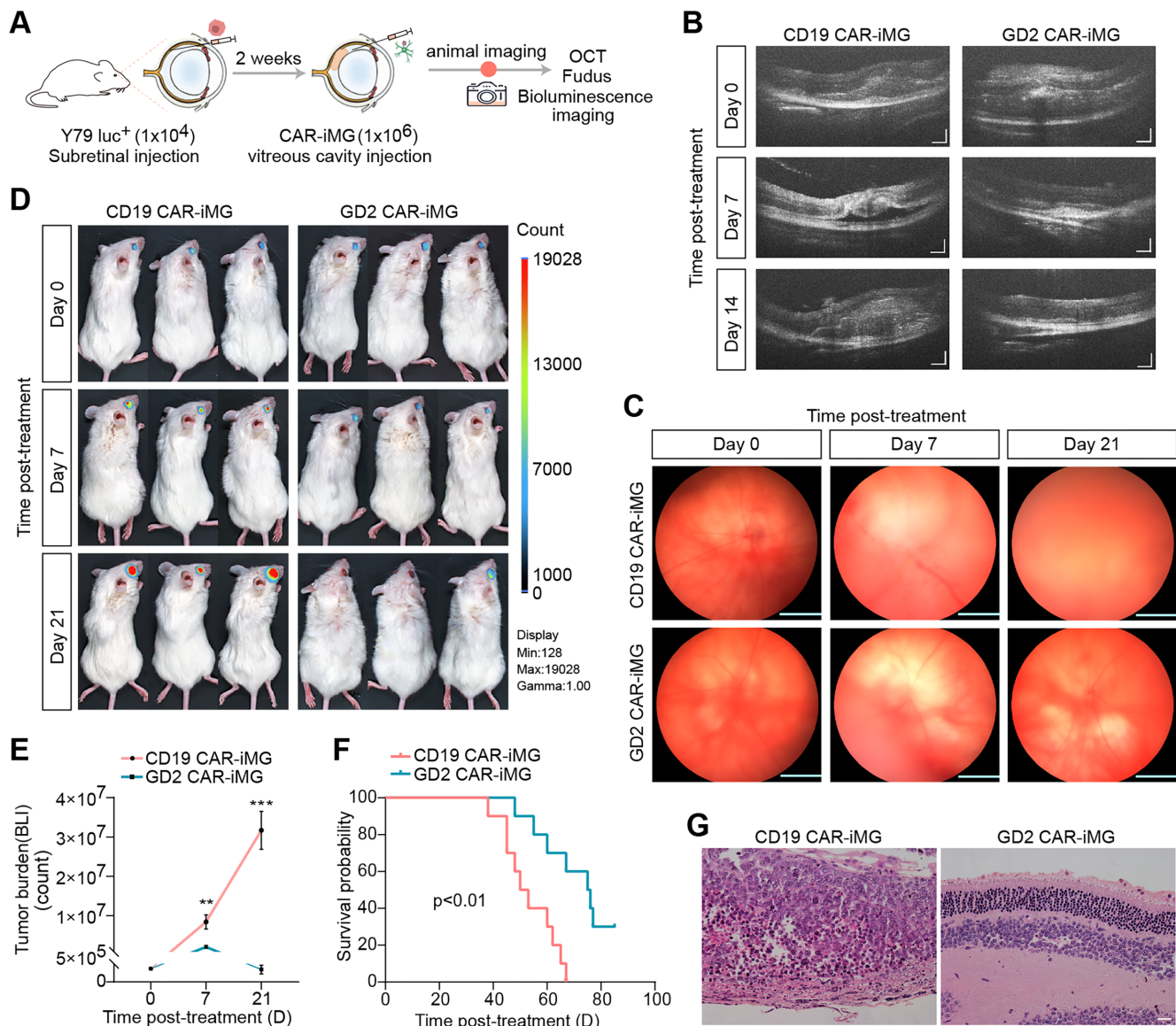


FIGURE 6. Efficacy and survival outcomes of GD2 CAR-iMG therapy in a retinoblastoma model. (A) An overview of the antitumor activity experiment using CD19 CAR-iMGs or GD2 CAR-iMGs in a Luc⁺ Y79 xenograft retinoblastoma model. (B) OCT images demonstrating reduced tumor volume and preservation of retinal layers in the GD2 CAR-iMG-treated mice. (C) Fundus photographs revealing decreased tumor size and altered vascular architecture in the GD2 CAR-iMG-treated mice. (D) Live animal imaging showing the time-dependent change of bioluminescence (BLI) from the retinoblastoma model after being treated by GD2 CAR-iMGs or CD19 CAR-iMGs for the indicated times. (Data on the same days were from the same experiment, and then the data were consolidated for each group) (n = 3 biologically independent samples per group). (E) The statistics of BLI from panel D for the indicated times. Data are shown as mean ± SD (n = 3). **P < 0.01, ***P < 0.001. (F) Kaplan-Meier curve representing the survival probability of the C-NKG mice after the indicated cell treatment (n = 10; log-rank test). (G) Hematoxylin and eosin staining of retinal sections, highlighting decreased tumor cell density and improved tissue architecture in the GD2 CAR-iMG-treated group after 7 days. Scale bar: 20 μm. Data are shown as mean ± SD (n = 3). **P < 0.01, ***P < 0.001.

tional specific markers for retinoblastoma or degeneration diseases will enhance the potential applications and efficacy of CAR-microglia treatments.

Acknowledgments

The authors thank Jing Yang, Shuning Sun, Dandan Shi, and Wen Wang for technical assistance, as well as Professor Qi Xie from Westlake University for helpful discussion.

Supported by the National Natural Science Foundation of China (82125007, 82101145, 92368206), National Key R&D Program of China (2022YFF1202901), Beijing Natural Science Founda-

tion (Z200014), Beijing Hospitals Authority Clinical Medicine Development of Special Funding Supports (ZLRK202503, DFL20220202), and the Beijing Municipal Public Welfare Development and Reform Pilot Project for Medical Research Institutes (PWD&RPP-MRI, JYY2023-6).

Data Availability Statements: The raw sequence data reported in this article have been deposited in the China National Center for Bioinformation/Beijing Institute of Genomics, Chinese Academy of Sciences (GSA-Human: HRA008917) and are publicly accessible at <https://ngdc.cnbc.ac.cn/gsa-human>.

Disclosure: J. Xu, None; Y. Gao, None; J. Wang, None; M. Lyv, None; Z. Zhang, None; Y. Cheng, None; Z.-B. Jin, None

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SUPPLEMENTARY MATERIAL

SUPPLEMENTARY MOVIE S1. A live imaging video demonstrated the GD2 CAR-microglia migrated (Green) towards Y79 retinoblastoma cells (Red) for 6 h.

SUPPLEMENTARY MOVIE S2. A live imaging video demonstrated the weak interaction between CD19 CAR-microglia (Green) and Y79 retinoblastoma cells (Red) for 6 h.

SUPPLEMENTARY MOVIE S3. A live imaging video demonstrated the promote tumor effect of CD19 CAR-iMG (Green) towards human retinoblastoma organoids (RBOs) (Brightfield) for 42 h.

SUPPLEMENTARY MOVIE S4. A live imaging video demonstrated the antitumor effect of GD2 CAR-iMG (Green) towards human retinoblastoma organoids (Brightfield) for 42 h.